Bryce Grant

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Robot manipulation and robot foundation models: training-free planning with runtime verification, mechanistic interpretability of VLAs, world-action models and transformers, and deployment on real arms.

Education

Case Western Reserve University — Cleveland, OH
Ph.D. Candidate, Electrical Engineering; advisor Peng (Edward) Wang — Expected May 2028
NSF Graduate Research Fellow (2025); NVIDIA Academic Grant (2025)

University of Kentucky — Lexington, KY
B.S. Computer Engineering and B.S. Electrical Engineering, Magna Cum Laude; Lexmark Scholar — May 2024

Publications (* equal contribution)

- **Not All Features Are Created Equal: A Mechanistic Study of Vision-Language-Action Models**. **B. Grant***, X. Zhao*, P. Wang. *NeurIPS 2026; ICLR 2026 Multimodal Intelligence Workshop (Oral)*.
- **Gluing Local Contexts into Global Meaning: A Sheaf-Theoretic Decomposition of Transformer Representations**. **B. Grant**, P. Wang. *NeurIPS 2026*.
- **Quaternion Approximate Networks for Enhanced Image Classification and Oriented Object Detection**. **B. Grant**, P. Wang. *IROS 2025 (Oral)*.
- **SPARK: Sequential Planning via Anchored Robotic Keypoints**. **B. Grant**, A. Rothenberg*, L. Senning*, P. Wang. *Under review, ICLR 2027; arXiv:2606.30613*.
- **TrianguLang: Geometry-Aware Semantic Consensus for Pose-Free 3D Localization**. **B. Grant**, A. Rothenberg, A. Banerjee, P. Wang. *Under review; arXiv:2603.08096*.

Research and Industry Experience

Case Western Reserve University, AISM Lab — Cleveland, OH
Graduate Research Assistant — Aug. 2024 – Present

- **SPARK (lead):** training-free manipulation; an LLM-planned behavior tree anchored to live 3D keypoints, verified per primitive with local recovery. 75.1% on LIBERO-PRO, 66.7% with objects moved 10 cm mid-task, 68% over 11 physical settings on UR10e, Franka FR3 and a bimanual rig.
- **VLA interpretability (co-lead):** 520+ sparse autoencoders over $\pi_{0.5}$, OpenVLA-OFT, X-VLA, SmolVLA and GR00T N1.5 ($\sim$10K A100-hours); activation injection shows policies bind motor programs to workspace coordinates (99.6% of $\pi_{0.5}$ episodes; 26/26 on a real UR10e). Released 7 TB of data and Action Atlas.
- **World-action models:** swapping only the imagined frames' keys and values redirects LingBot-VA (39/40) and DreamZero (40/40) as a new instruction would; the target is read at the last denoising steps, and masking the imagined future lowers success on all eight WAMs tested.
- **Sheaf interpretability:** a training-free cellular sheaf on paraphrase graphs; its 20-dim kernel beats LEACE by +17.1 pp on held-out fact retrieval (8 LLMs), and its complement localizes causal influence (9 models, 124M–13B).
- Research with AI coding agents under hard validation: agent-written audits caught a leak in my own benchmark harness (disclosed and rerun); a 1,300-test CI gates the SPARK codebase.

Mercor — Remote
Applied AI Engineer Intern — Aug. 2025 – Aug. 2026

- Built RL environments, synthetic-data pipelines and trajectory-annotation tools for coding-agent post-training (RLHF/SFT) programs; set acceptance criteria for agent runs.

HP — Vancouver, WA
PhD ML Intern — May – Aug. 2024

- Benchmarked RAG designs (chunking, BM25 vs. FAISS, query expansion; NDCG) and shipped an LLM support agent; LSTM + STL anomaly detection on AWS cloud spend.

Skills and Service

- **Robots & sim:** UR10e/UR5e, Franka FR3 (incl. bimanual), Unitree G1/Go2, FlexiTac tactile; MuJoCo, MuJoCo-Warp, Isaac Sim/Lab, ROS 2, cuRobo, LIBERO, RoboTwin 2.0
- **Models:** VLAs ($\pi_{0.5}$, OpenVLA-OFT, GR00T, SmolVLA), world-action models (Cosmos Policy, DreamZero), flow matching, sparse autoencoders, SAM3, Depth Anything 3
- **Engineering:** Python, C/C++, CUDA, Warp, PyTorch, JAX, LeRobot, vLLM; Slurm (Delta/Anvil, $\sim$10K A100-hours), Docker, Linux
- Reviewer: CoRL 2026, ICLR 2026–27, ECCV 2026, IROS 2025–26.